

Improving disulfide connectivity prediction with sequential distance between oxidized cysteines

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Chi-Hung Tsai , Bo-Juen Chen , Chen-hsiung Chan,
Hsuan-Liang Liu and Cheng-Yan Kao

Presenter: Wei-Chun Chung

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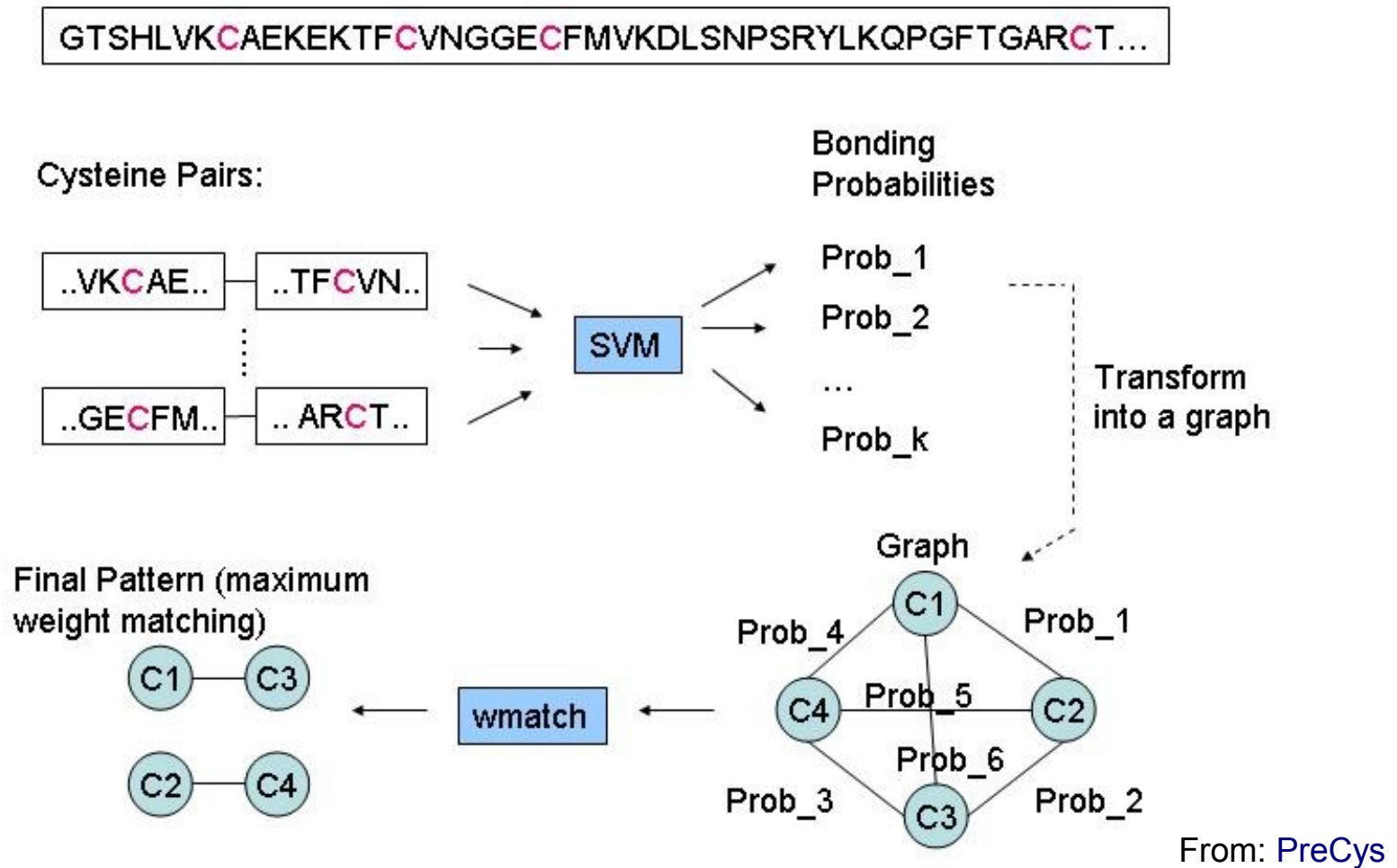
Abstract (1/2)

- Predicting disulfide connectivity precisely helps towards the solution of protein structure prediction. In this study, a descriptor derived from the sequential distance between oxidized cysteines (denoted as DOC) is proposed. An approach using support vector machine (SVM) method based on weighted graph matching was further developed to predict the disulfide connectivity pattern in proteins.

Abstract (2/2)

- When DOC was applied, prediction accuracy of 63% for our SVM models could be achieved, which is significantly higher than those obtained from previous approaches. The results show that using the non-local descriptor DOC coupled with local sequence profiles significantly improves the prediction accuracy. These improvements demonstrate that DOC, with a proper scaling scheme, is an effective feature for the prediction of disulfide connectivity. The method developed in this work is available at the web server PreCys (prediction of cys–cys linkages of proteins).

Methodology



Data Encoding

- Local sequence profile: by PSI-BLAST
 - A sliding window center at Cystine
- Linear DOC: $DOC(i, j) = \|i - j\|$
 - DOC_L : DOC normalized with the protein sequence length L .
 - DOC_{max} : DOC normalized with the maximum value of the whole dataset.
 - DOC_{log} : DOC values normalized with the logarithm function

PSSM (1/3)

- Position Specific Scoring Matrix

The *score* is derived from the ratio of the observed to the expected frequencies. More precisely, the logarithm of this ratio is taken and referred to as the *log-likelihood ratio*:

$$Score_{ij} = \log\left(\frac{f'_{ij}}{q_i}\right)$$

where $Score_{ij}$ is the score for residue i at position j , f'_{ij} is the relative frequency for a residue i at position j (corrected with pseudo-counts) and q_i is the expected relative frequency of residue i in a random sequence.

PSSM (2/3)

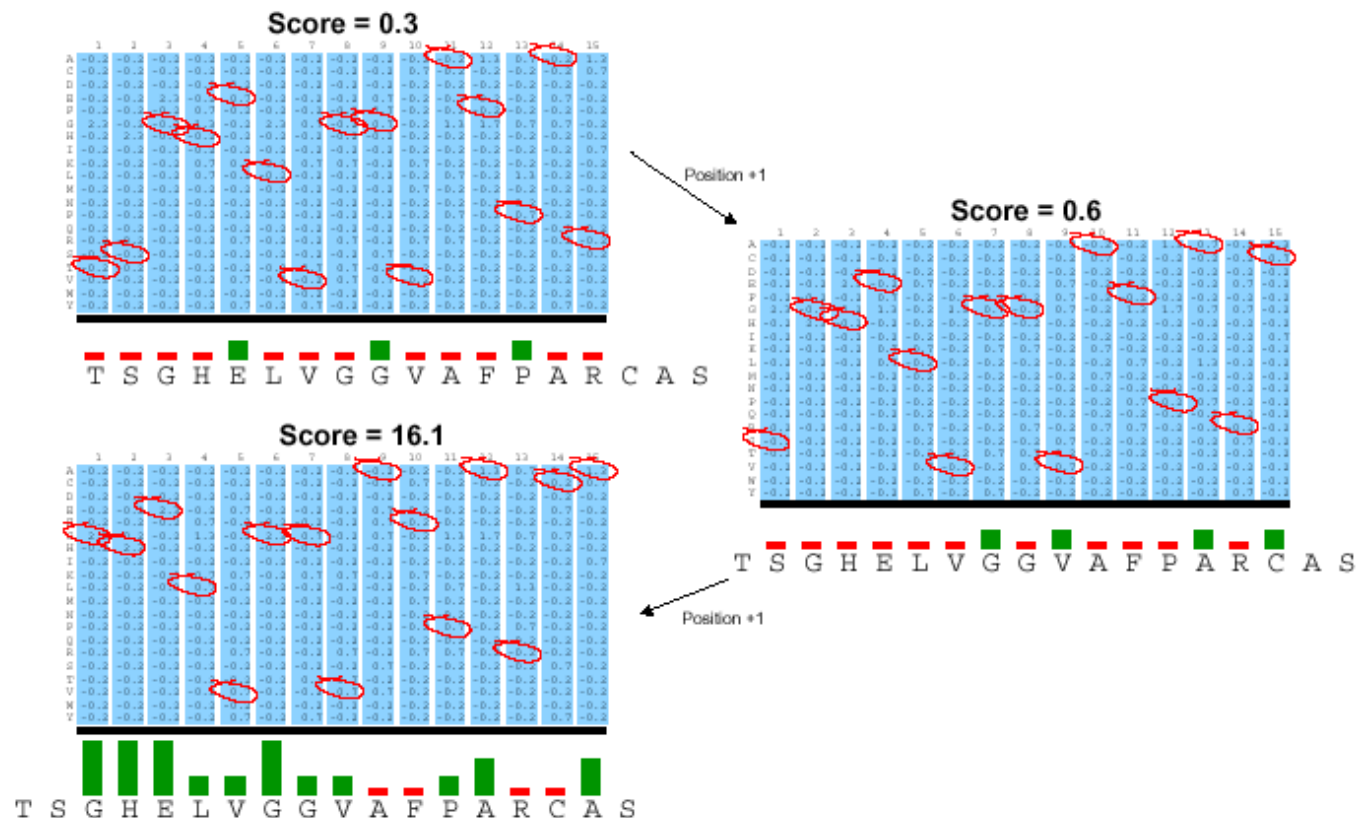
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	0	0	0	0	0	0	0	0	0	0	0	2	1	0	2
C	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1
D	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	0	0	5	0	1	0	0	0	1	0	0	0	0	1	0
F	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0
G	5	0	0	2	0	5	1	0	1	0	2	3	1	1	0
H	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0
I	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
K	0	0	0	1	1	0	1	1	0	1	0	0	0	0	0
L	0	0	0	1	0	0	0	0	0	0	1	0	2	0	0
M	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0
Q	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R	0	0	0	0	1	0	0	1	0	1	1	0	0	0	0
S	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0
T	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0
V	0	0	0	0	1	0	0	1	1	0	0	0	0	0	0
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Y	0	0	0	0	1	0	1	0	0	0	0	0	0	1	0

PSSM (3/3)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	1.3	0.7	-0.2	1.3
C	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	0.7	-0.2	-0.2	-0.2	-0.2	0.7
D	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
E	-0.2	-0.2	2.3	-0.2	0.7	-0.2	-0.2	-0.2	0.7	-0.2	-0.2	-0.2	-0.2	0.7	-0.2
F	-0.2	-0.2	-0.2	0.7	-0.2	-0.2	-0.2	-0.2	0.7	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
G	2.3	-0.2	-0.2	1.3	-0.2	2.3	0.7	-0.2	0.7	-0.2	1.3	1.7	0.7	0.7	-0.2
H	-0.2	2.3	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
I	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	0.7
K	-0.2	-0.2	-0.2	0.7	0.7	-0.2	0.7	0.7	-0.2	0.7	-0.2	-0.2	-0.2	-0.2	-0.2
L	-0.2	-0.2	-0.2	0.7	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	0.7	-0.2	1.3	-0.2	-0.2
M	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	0.7	-0.2	-0.2	-0.2	-0.2	-0.2
N	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
P	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	0.7	-0.2	0.7	-0.2	-0.2
Q	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
R	-0.2	-0.2	-0.2	-0.2	0.7	-0.2	-0.2	0.7	-0.2	0.7	0.7	-0.2	-0.2	-0.2	-0.2
S	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	0.7	-0.2	-0.2	-0.2	-0.2	0.7	-0.2
T	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	0.7	0.7	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
V	-0.2	-0.2	-0.2	-0.2	0.7	-0.2	-0.2	0.7	0.7	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
W	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Y	-0.2	-0.2	-0.2	-0.2	0.7	-0.2	0.7	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	0.7	-0.2

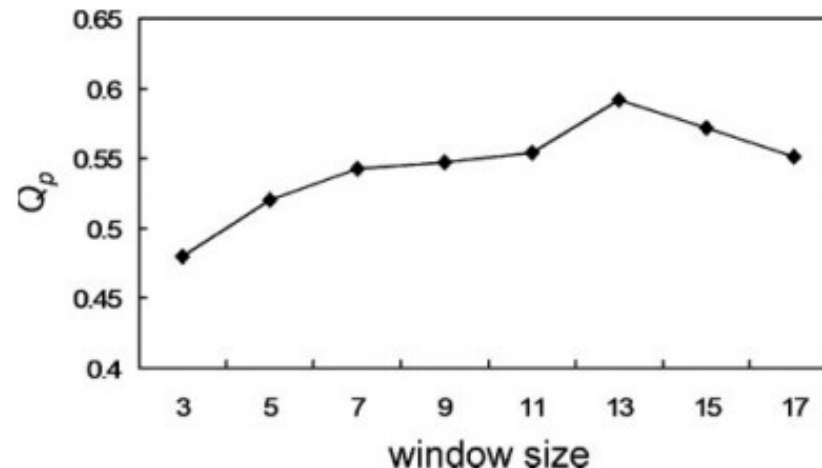
PSI-BLAST

- The PSSM is applied as a *sliding window* along the subject sequence:
 - At every position, a PSSM score is calculated by summing the scores of all columns;
 - The highest scoring position is reported.

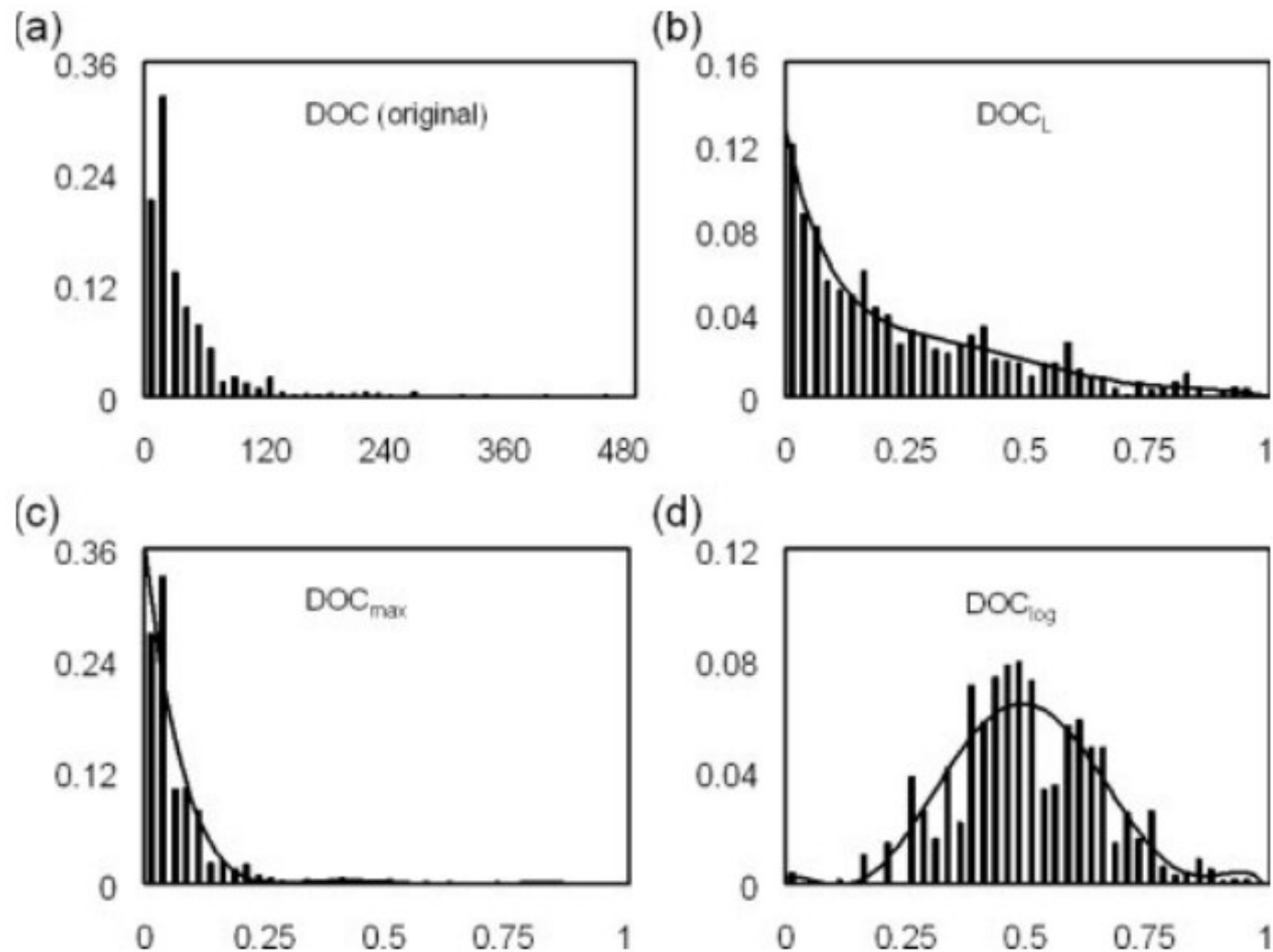


Implementation (1/2)

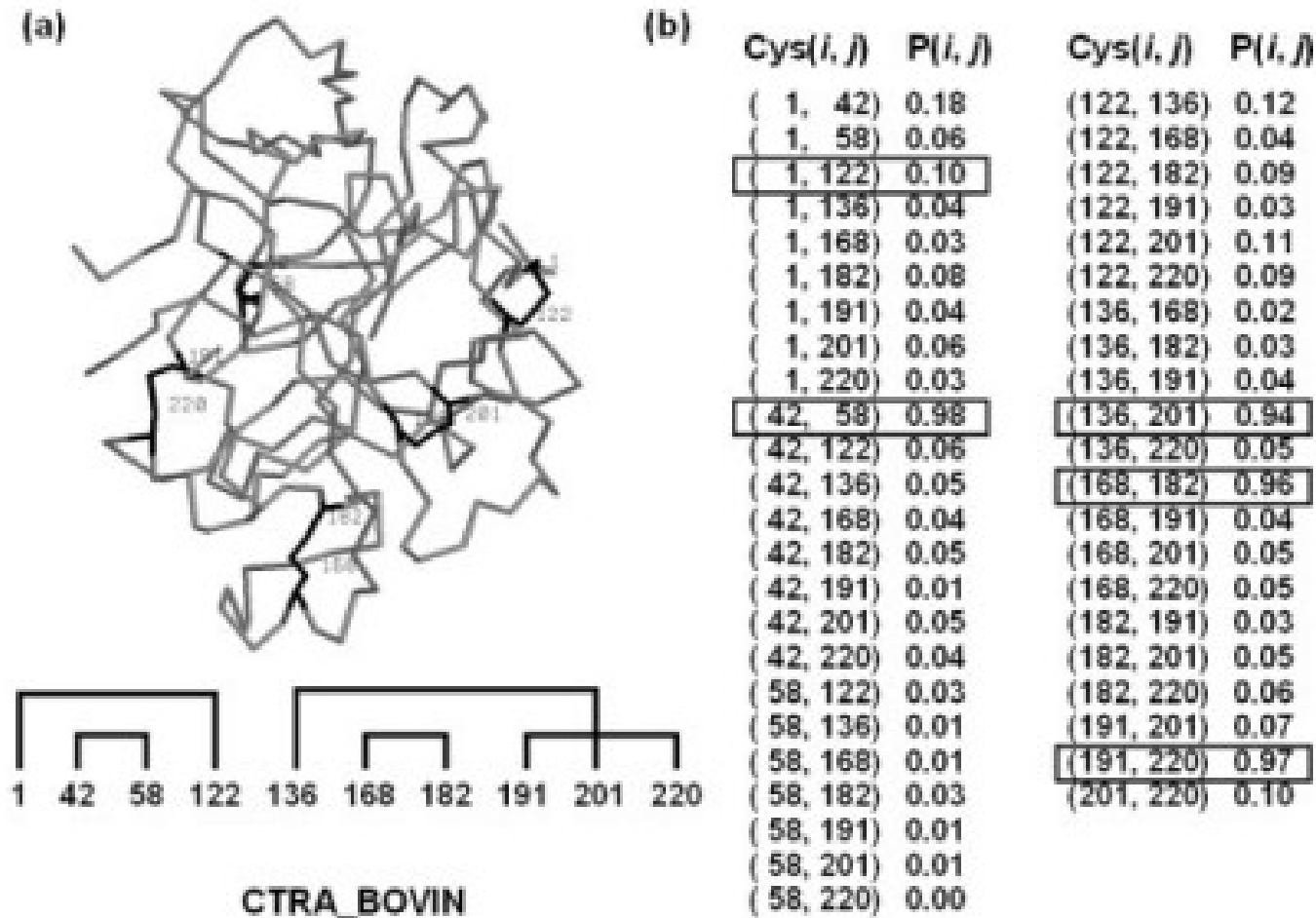
- Dataset: SP-39
 - 4-fold cross-validation
- Evaluation Criteria
 - $Q_p = C_p / T_p$ (protein based)
 - $Q_c = C_c / T_c$ (connection based)
- Window size: 13



Implementation (2/2)



Sample



Result

Methods	$B = 2$		$B = 3$		$B = 4$		$B = 5$		$B = 2-5$	
	Q_p (%)	Q_c (%)	Q_p (%)	Q_c (%)	Q_p (%)	Q_c (%)	Q_p (%)	Q_c (%)	Q_p (%)	Q_c (%)
MC graph-matching ^a	56	56	21	36	17	37	2	21	29	38
NN graph-matching ^b	68	68	22	37	20	37	2	26	34	42
BiRnn-2 profile ^c	73	73	41	51	24	37	13	30	44	49
2D-Rnn profile ^d	74	74	51	61	27	44	11	41	49	56
dNN2 ^e	62	—	40	—	55	—	26	—	49	—
CSP	72	72	54	66	33	50	18	36	52	58
SVM profile	76	76	53	62	48	62	44	60	59	65
SVM profile + DOC_{log}	79	79	53	62	55	70	58	71	63	70

Introduction

Methodology

I. pair-wise models

II. two-level models

Service

Link

Contacts

Oxidized Cysteines: 27 42 47 93 99 140

Bonding Pattern

Cysteine i	Position	Cysteine j	Position	Distance	Bonding Prob.
1	27	2	42	14	0.03884
1	27	3	47	19	0.03881
1	27	4	93	65	0.03888
1	27	5	99	71	0.03885
1	27	6	140	112	0.96631
2	42	3	47	4	0.96623
2	42	4	93	50	0.03885
2	42	5	99	56	0.03888
2	42	6	140	97	0.03883
3	47	4	93	45	0.03741
3	47	5	99	51	0.03885
3	47	6	140	92	0.03883
4	93	5	99	5	0.96629
4	93	6	140	46	0.03886
5	99	6	140	40	0.03881

Final pattern: (1,6) (2,3) (4,5)

Updated on May 17, 2006